EXP.NO:02

PROBLEM 2 : FIND THE TIME COMPLEXITY USING COUNTER METHOD

PROBLEM STATEMENT:

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

EXP.NO:02

CODE:

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n, count=0;
    scanf("%d", &n);

    if(n==1){
        count++;
    }
    else{
        count++;

        for(int i=1;i<=n;i++){


            count++;
            count++;
            count++;
            count++;

            count++;
        }
        count++;

    }
    printf("%d", count);
    return 0;
}
```

OUTPUT:

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.